

Po-Chun Chien

Curriculum Vitae

2026-05-26

Coordinates

 Affiliation:	Institute of Informatics, LMU Munich, Oettingenstr. 67, 80538 Munich, Germany	 Webpage:	www.sosy-lab.org/people/chien/
		 Email:	po-chun.chien@sosy.ifi.lmu.de
		 ORCID:	0000-0001-5139-5178

Research Interests

- Formal Methods: New algorithms and tools for the verification of computational systems
- Electronic Design Automation: Logic synthesis and optimization of hardware digital circuits
- Machine Learning: Application to formal verification and logic synthesis

Skills

- Programming: C/C++, Java, Python, Verilog HDL, Bash script, Cadence SKILL, MATLAB
- Natural Language: Mandarin Chinese (native), English (fluent)

Education

- 2021–2026 (expected) Ph.D. in Informatics (Computer Science), LMU Munich, Germany
- Advisor: Prof. Dirk Beyer
 - Working on Bridging Automated Software and Hardware Verification
 - Funded by German Research Foundation (DFG): [ConVeY](#) and [Bridge](#)
- 03–07 2025 Research visit at Stanford University, USA
- Host: Prof. Clark Barrett
 - Working on Word-Level Safety and Liveness Model Checking
 - Funded by DAAD Research Fellowship and DFG Bridge
- 07–10 2024 Research visit at University of Freiburg, Germany
- Host: Prof. Armin Biere
 - Working on Encoding Software Verification Tasks as Hardware Circuits
 - Funded by DFG ConVeY
- 2018 – 2020 M.Sc. in Electronics Engineering, National Taiwan University (NTU), Taiwan
- Advisor: Prof. Jie-Hong Roland Jiang
 - Thesis: Circuit Folding: From Combinational to Sequential Circuits
 - Overall GPA: 4.22 / 4.30
- 2015 – 2018 B.Sc. in Electrical Engineering, National Taiwan University, Taiwan
- Overall GPA: 4.16 / 4.30 (top 5 %)

Academic Employment

- since 2021 Doctoral Researcher (advisor: Prof. Dirk Beyer) at
LMU Munich, Munich, Germany
- 2018 – 2021 Research and Teaching Assistant (advisor: Prof. Jie-Hong Roland Jiang) at
National Taiwan University, Taipei, Taiwan

Industrial Employment

- 2018 Summer Intern at MediaTek, Hsinchu, Taiwan

Honors & Awards

Academic Awards

2026	Best Tool Paper Award at the International Symposium on Formal Methods (FM)
2026	ETAPS Distinguished Paper Award at the Intl. Joint Conferences On Theory and Practice of Software
2024	ACM SIGSOFT Distinguished Paper Award and Best Artifact Award at the ACM Intl. Conference on the Foundations of Software Engineering (FSE)
2024	Distinguished Artifact Award at the Intl. Conference on Tools and Algorithms for the Construction and Analysis of System (TACAS)
2024	Best Paper Award at the Intl. Symposium on Model Checking Software (SPIN)
2020	Chinese Institute of Electrical Engineering Thesis Award
2020	Graduate Institute of Electronics Engineering Outstanding Master's Thesis Award
2020	Institute of Information & Computing Machinery Master's Thesis Excellence Award
2020	Lam Research Master's Thesis Excellence Award at NTU
2016 – 2017	NTU Presidential Award (2 semesters)

Competition Awards

2025	1st place in Word-Level Arrays Track of Hardware Model Checking Comp. (HWMCC)
2023, 2024	2nd place overall in the Intl. Competition on Software Verification (SV-COMP)
2022	3rd place overall in the Intl. Competition on Software Verification (SV-COMP)
2021	2nd place in the programming contest of the Intl. Workshop on Logic & Synthesis
2019	1st place in CADathlon at the Intl. Conference on Computer-Aided Design (ICCAD)
2019	1st place in the Formosa Grand Challenge "Taking with AI" (domestic)

Scholarships

2025	DAAD Research Fellowship for exchange abroad (5 months)
2018 – 2020	NTU-GIEE Scholarship (4 semesters)
2017 – 2018	TSMC-NTU Scholarship (2 semesters)

Publications

My publications can be found on [DBLP](#) and [Google Scholar](#). Five selected ones are listed below.

1. Áron Ricardo Perez-Lopez, **Po-Chun Chien**, Florian Lonsing, Samantha Archer, Ahmed Irfan, and Clark Barrett. Pono 2.0: A versatile SMT-based model checker for safety and liveness (long tool paper). In *Proceedings of the International Symposium on Formal Methods (FM)*, LNCS 16557, pages 3–26. Springer, 2026. doi: [10.1007/978-3-032-26220-2_1](#). [**Best Tool Paper Award**].
2. Salih Ates, Dirk Beyer, **Po-Chun Chien**, and Nian-Ze Lee. Bridging hardware and software analysis with Btor2C: A word-level-circuit-to-C translator (extended version). *International Journal on Software Tools for Technology Transfer (STTT)*, 2026. doi: [10.1007/s10009-026-00847-z](#). [**Special Issue for TACAS 2023**].
3. Dirk Beyer, **Po-Chun Chien**, Bo-Yuan Huang, Nian-Ze Lee, , and Thomas Lemberger. A case study in firmware verification: Applying formal methods to Intel TDX Module. In *Proceedings of the International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*, LNCS 16506, pages 42–64. Springer, 2026. doi: [10.1007/978-3-032-22749-2_3](#). [**ETAPS Distinguished Paper Award**].
4. Dirk Beyer, **Po-Chun Chien**, Marek Jankola, and Nian-Ze Lee. A transferability study of interpolation-based hardware model checking for software verification. *Proceedings of the ACM on Software Engineering*, 1(FSE), 2024. doi: [10.1145/3660797](#). [**ACM SIGSOFT Distinguished Paper Award**].
5. Dirk Beyer, **Po-Chun Chien**, and Nian-Ze Lee. Augmenting interpolation-based model checking with auxiliary invariants. In *Proceedings of the International Symposium on Model Checking Software (SPIN)*, LNCS 14624, pages 227–247. Springer, 2024. doi: [10.1007/978-3-031-66149-5_13](#). [**Best Paper Award**].

Software Projects

- CPV: Circuit-based program verification
<https://gitlab.com/sosy-lab/software/cpv>
Role: principle designer, maintainer, and developer
- BTOR2-CERT: A certifying hardware-verification framework using software analyzers
<https://gitlab.com/sosy-lab/software/btor2-cert>
 - BTOR2C: A translator from word-level circuits to C programs
<https://gitlab.com/sosy-lab/software/btor2c>
 - BTOR2-VAL: A witness validator for word-level hardware model checking
<https://gitlab.com/sosy-lab/software/btor2-val>Role: principle designer, maintainer, and developer
- PONO: A flexible and extensible SMT-based model checker
<https://github.com/stanford-centaur/pono>
Role: developer
- SMT-SWITCH: A solver-agnostic C++ API for SMT solving
<https://github.com/stanford-centaur/smt-switch>
Role: developer
- MOXICHECKER: An extensible model checker for MoXI
<https://gitlab.com/sosy-lab/software/moxichecker>
Role: principle designer, maintainer, and developer
- BTOR2-SELECT: Machine-learning-based algorithm selection for hardware model checking
<https://gitlab.com/sosy-lab/software/btor2-select>
Role: maintainer and developer
- HARNESSFORGE: Automated tooling for generating verification tasks from source code
<https://gitlab.com/sosy-lab/software/harnessforge>
Role: developer
- CPACHECKER: A configurable software-verification framework
<https://cpachecker.sosy-lab.org/>
Role: developer
- BENCHCLOUD: A cloud platform for scalable performance benchmarking
<https://gitlab.com/sosy-lab/software/benchcloud>
Role: maintainer and developer
- BENCHEXEC: A framework for reliable benchmarking and resource measurement
<https://github.com/sosy-lab/benchexec>
Role: contributor
- EXT-FOLDING: A circuit-folding interface for the logic synthesis system [ABC](#)
<https://github.com/NTU-ALComLab/ext-folding>
Role: principle designer and developer

Student Mentoring

2025–2026	Qifan Yang, Master student, LMU Munich Research assistant for BENCHCLOUD
2025	Luiza Pires Ribeiro, Bachelor student, Stanford University CURIS Summer Internship Program Topic: Examining Witness Validity of Pono
2024	Zhengyang (John) Lu, PhD student, University of Waterloo Google Summer of Code Topic: Adaptive Algorithm Selection for Btor2 Verification Tasks
2024	Diego Salgado Esparza, Bachelor student, LMU Munich Research assistant for BENCHCLOUD
2024	Salih Ates, Mater student, LMU Munich Research assistant for BTOR2C and MOXICHECKER
2023	Zsófia Ádám, Exchange student, Budapest University of Technology and Economics Erasmus Program Topic: Correctness Witness Validation for Programs Translated from Hardware Models

2023	Nils Sirrenberg, Bachelor student, LMU Munich LMU Bachelor's Thesis Topic: Certifying Software Violation Witnesses for Hardware Verification Tasks via Simulation-Based Validation
2023	Jia Sun, Bachelor student, Kyoto University Google Summer of Code Topic: Reverse Program Synthesis for Backward Reachability Analysis
2023	Salih Ates, Bachelor student, LMU Munich LMU Bachelor's Thesis Topic: Improving Array Encoding in Hardware-to-Software Translation
2023	Zhu Yang, Master Student, LMU Munich Research assistant for BENCHCLOUD

Teaching Experience

I have been a teaching assistant for the following courses.

Graduate Courses

Summer 2026	Software Engineering Internship (instructor: Prof. Dirk Beyer)
Summer 2024	Software Engineering Internship (instructor: Prof. Dirk Beyer)
Summer 2023	Software Engineering Internship (instructor: Prof. Dirk Beyer)
Summer 2022	Software Engineering Internship (instructor: Prof. Dirk Beyer)
Winter 2021	Software Verification (instructor: Prof. Dirk Beyer)
Fall 2018	Deep Learning for Human Language Processing (instructors: Prof. Hung-Yi Lee and Prof. Yun-Nung Chen)
Spring 2018	Advanced Deep Learning (instructors: Prof. Hung-Yi Lee and Prof. Yun-Nung Chen)

Graduate Seminars

Winter 2025	Reproducibility of Software Engineering Research (instructor: Dr. Stefan Winter)
Winter 2024	Deductive Software Verification (instructor: Prof. Gidon Ernst)
Summer 2024	Algorithms for Model Checking (instructor: Prof. Dirk Beyer)
Winter 2023	Algorithms for Model Checking (instructor: Prof. Dirk Beyer)

Undergraduate Courses

Summer 2021	Software Development Internship (instructor: Prof. Dirk Beyer)
Spring 2020	Introduction to Electronic Design Automation (instructor: Prof. Jie-Hong Roland Jiang)
Spring 2019	Introduction to Electronic Design Automation (instructor: Prof. Jie-Hong Roland Jiang)

Undergraduate Seminars

Winter 2022	Software Verification: Tools and Techniques (instructor: Prof. Dirk Beyer)
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Professional Activities

Conference / Workshop / Competition Organization

2026	Organization committee (infrastructure) of SV-COMP Student Volunteer at the International Conference on the Art, Science, and Engineering of Programming (·Programming·)
2025	Organization committee (infrastructure) of SV-COMP
2024	Co-organizer of the International Workshop on CPAchecker
2022	Student Volunteer at the European Joint Conferences on Theory and Practice of Software (ETAPS)

Journal Referee

2024 IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems

Conference Referee (co/sub-reviewer)

2026 International Conference on the Foundations of Software Engineering (FSE)
International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS)
International Symposium on Formal Methods (FM)
International Conference on Automated Software Engineering (ASE)

2025 International Conference on Computer Aided Verification (CAV)
International Conference on the Foundations of Software Engineering (FSE)
International Conference on Automated Software Engineering (ASE)
Asia and South Pacific Design Automation Conference (ASP-DAC)

2024 International Conference on the Foundations of Software Engineering (FSE)
International Conference on Software Engineering (ICSE)
International Symposium on Formal Methods (FM)

2023 International Conference on Computer Design (ICCD)
International Conference on Computer Aided Verification (CAV)
International Conference on Automated Software Engineering (ASE)

2022 NASA Formal Methods Symposium (NFM)
International Symposium on Software Reliability Engineering (ISSRE)
Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)

2021 International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI)

2020 Design Automation Conference (DAC)
International Conference on Computer-Aided Design (ICCAD)

Artifact Evaluation Committee

2026 International Symposium on Formal Methods (FM)

2025 International Conference on Computer Aided Verification (CAV)

2024 International Conference on Computer Aided Verification (CAV)
International Symposium on Automated Technology for Verification and Analysis (ATVA)

Google Summer of Code

2023, 2024 Org admin and mentor for Software and Computational Systems Lab at LMU Munich

References

Available upon request.